

Whatever Happens in Brno, Definitely Does Not Stay in Brno!

**Report on Computer Vision in Archaeology Training School of ATRIUM
Project**

(4-18 September 2026)



Fig.1: Cheese! (<https://arup-cas.github.io/atrium-school-ml/report.html#photos>)

Onur Altun

University of Cologne

Digital and Computational Archaeology

A Word of Thanks

From 14 to 18 September 2026, I had the wonderful opportunity to participate in the Computer Vision in Archaeology Training School in Brno, Czech Republic. This five-day program was hosted by the Archaeological Information System of the Czech Republic at the Institute of Archaeology of the Czech Academy of Sciences. It was organized as part of the ATRIUM Project and co-organized with the MAIA COST Action.

Coming from a background in Classical Archaeology at Mimar Sinan Fine Arts University and currently continuing my work at the University of Cologne, this training was a very special and important experience for me. It gave me the chance to meet, talk, laugh and share ideas with many different researchers and experts. Personally, these conversations helped me gain very valuable insights, especially regarding the "slow science" and "slow archaeology" perspectives that I am currently focusing on in my own studies.

I would like to sincerely thank all the host institutions, the amazing instructors and my fellow participants who put so much hard work and dedication into making this beautiful event happen. Finally, I owe a very special thank you to my Head of Department, Prof. Dr. Eleftheria Paliou, who first told me about this great opportunity and made it possible for me to be there.

Program

The first day started with a very warm welcome and introductions. Petr Pajdla gave an interactive presentation on the basics of computer vision in archaeology, showing us the typical workflows and limitations through real case studies. For the practical part, we work with CLIP for zero-shot classification. This session was both educational and really funny. We got to see exactly how well-trained the AI is when it confidently decided that the rock band Led Zeppelin simply does not belong in the same category as normal humans! Afterwards, Ronald Harasim shared useful tips on coding with AI agents and Filip Hájek covered Python basics, so everyone was on the same page. We wrapped up this busy day with a relaxing ice-breaker event on the terrace. It was the perfect way to chat, enjoy a few drinks and build a friendly collaborative atmosphere for the rest of the week.

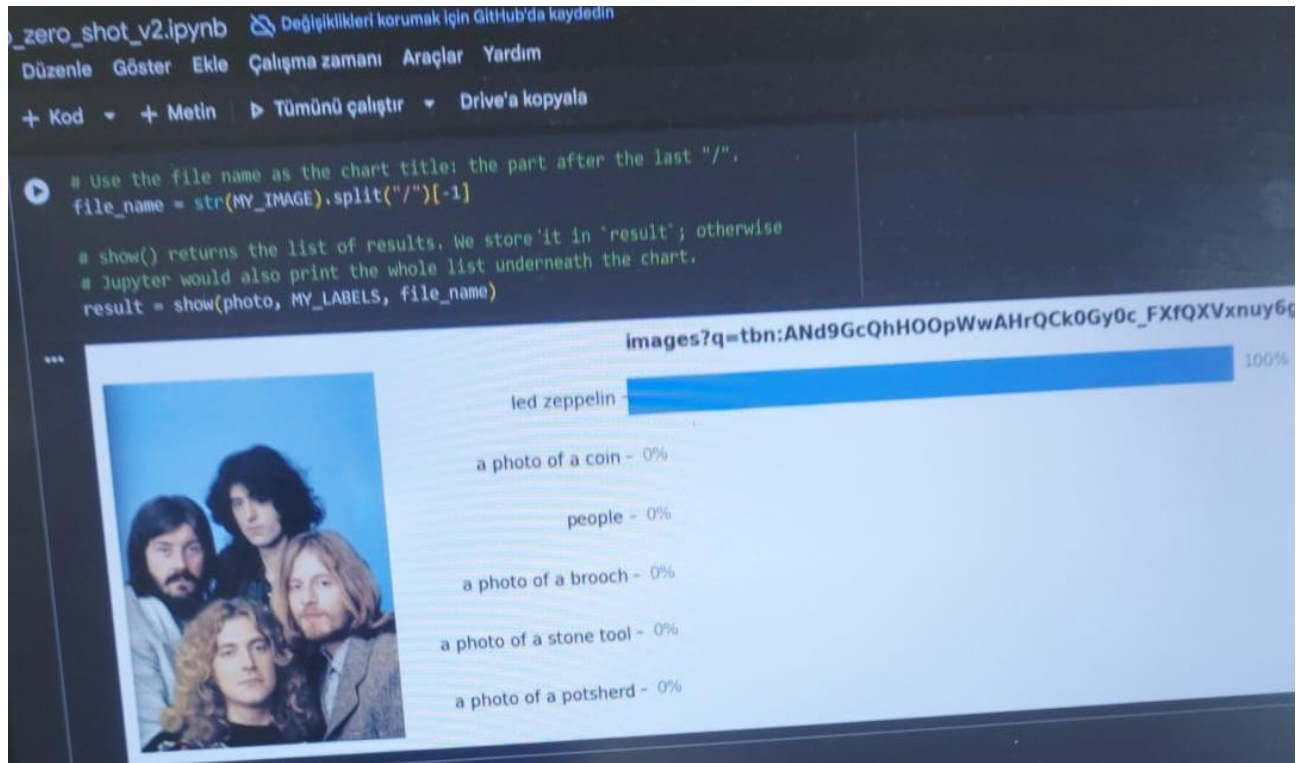


Fig. 2: When the well-trained AI meets with Led Zeppelin.

The second day focused on archaeological artifact images and dataset preparation. We learned how to annotate images using tools like CVAT to train predictive algorithms. In the afternoon, participants had the chance to present their own research projects, which was a great way to learn about everyone's work. The day ended beautifully with a guided walking tour of Brno, where we explored the city's historic sights and enjoyed some local places together.



Fig. 3: Brno (<https://arup-cas.github.io/atrium-school-ml/report.html#photos>)

On the next day, we shifted our focus to microscopic images, specifically looking at use-wear traces. We practiced pre-processing techniques like standardizing brightness and contrast and used OpenCV for image masking. During the break, we were given a fascinating tour of the institute's laboratories to see how physical artifacts are handled before they become digital data.

We wrapped up the day with a lovely social event at the institute, tasting local food and snacks. The fourth day was divided into two exciting topics: satellite imagery and rock art. We explored remote sensing techniques to extract geospatial data and detect archaeological features and then learned how to enhance rock art images using algorithms like CLAHE to identify patterns. Later, we visited the institute's archives, which gave us a deeper understanding of the research infrastructure available.

For the final day, we brought everything together by working with a dataset of coins. We also dove into cutting-edge topics like Large Language Models and vision-language models, having very important discussions about model bias, interpretability and the limits of "black-box" AI.



Fig. 4 (<https://arup-cas.github.io/atrium-school-ml/report.html#photos>)

CONCLUSION

To conclude, I think it is time to update a famous saying: whatever happens in Brno, definitely does not stay in Brno! The moments we shared, the laughs we had and the connections we created are now a permanent part of our memories. These experiences certainly won't remain confined to just one location; we will carry them with us wherever our research takes us next. Thank you to everyone who made this week so unforgettable, it is for sure that what we built here is just the beginning.

